CodInClub
Powered By BridgeLabz

C#: Class and Object

1. What is a Class?

A class is a blueprint or template that defines the structure (attributes) and behavior (methods) of objects. It provides a way to group related data and functions together.

- A class doesn't consume memory until objects are created from it.
- It is defined using the class keyword.

Syntax:

```
Java
class ClassName {
    // Attributes (fields)
    DataType attributeName;

    // Methods (behavior)
    returnType MethodName(parameters) {
        // Method body
    }
}
```

Example:

```
Java
class Car {
    // Attributes
    public string Brand;
    public int Speed;

    // Constructor
    public Car(string brand, int speed) {
        this.Brand = brand;
        this.Speed = speed;
    }

    // Method
    public void DisplayDetails() {
        Console.WriteLine("Brand: " + Brand + ", Speed: " + Speed + "
km/h");
    }
}
```

2. What is an Object?

An object is an instance of a class that represents a real-world entity. It has:

- **State:** Defined by the values of its attributes.
- **Behavior:** Defined by the methods of the class.

Syntax:

ClassName objectName = new ClassName(parameters);

Example:

```java
class Program {
    static void Main(string[] args) {
        // Creating an object of the Car class
        Car myCar = new Car("Tesla", 200);
        myCar.DisplayDetails();  // Output: Brand: Tesla, Speed: 200 km/h
    }
}
```

Class vs. Object

Aspect	**Class**	**Object**
Definition	A blueprint or template for creating objects.	An instance of a class.
Memory	Does not occupy memory.	Occupies memory when created.
Usage	Defines attributes and behaviors.	Represents specific attributes and behaviors.
Example	class Car {}	Car myCar = new Car();

Key Concepts

- **Attributes (Fields):** Represent the data or state of the object.
  - Example: public string Brand;, public int Speed;
- **Methods:** Define the behavior or actions of the object.
  - Example: public void DisplayDetails()
- **Constructor:** A special method used to initialize the attributes of an object.
  - Example:

```
public Car(string brand, int speed) {
    this.Brand = brand;
    this.Speed = speed;
}
```

- **Memory Allocation:** When an object is created, memory is allocated for its attributes in the heap.

Best Practices

- Use meaningful class names:

```
class Car {}  // Good
class C {}    // Bad
```

- Encapsulate data by marking attributes as private and providing getter and setter methods.
- Follow proper naming conventions for attributes and methods (camelCase).
- Always provide a default or parameterized constructor.

CodInClub

Powered By BridgeLabz

Programming Practice

Program 1: Displaying the Details of a Student

```
Java
class Student {
    public string Name;
    public int RollNumber;
    public double Marks;

    // Constructor
    public Student(string name, int rollNumber, double marks) {
        this.Name = name;
        this.RollNumber = rollNumber;
        this.Marks = marks;
    }

    // Method to display student details
    public void DisplayDetails() {
        Console.WriteLine("Name: " + Name + ", Roll Number: " + RollNumber +
", Marks: " + Marks);
    }
}

class Program {
    static void Main(string[] args) {
        Student student = new Student("Alice", 101, 89.5);
        student.DisplayDetails();  // Output: Name: Alice, Roll Number: 101,
Marks: 89.5
    }
}
```

Program 2: Travel Details

```
Java
class TravelDetails {
    public string FromCity, ToCity;
    public double Distance;

    // Constructor
    public TravelDetails(string fromCity, string toCity, double distance) {
```

```
        this.FromCity = fromCity;
        this.ToCity = toCity;
        this.Distance = distance;
    }

    // Method to display travel information
    public void DisplayTravelInfo() {
        Console.WriteLine("Traveling from " + FromCity + " to " + ToCity + " covers " + Distance + " km.");
    }
}

class Program {
    static void Main(string[] args) {
        TravelDetails travel = new TravelDetails("Chennai", "Bangalore", 345.6);
        travel.DisplayTravelInfo();  // Output: Traveling from Chennai to Bangalore covers 345.6 km.
    }
}
```